Ethical Reconnaissance Report OSINT and Nmap Mapping

- **Target Entity:** Axia Technology & Innovation Hub (`axiahub.ng`)
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- **Course / Program:** Cybersecurity SIWES Training Programme

1. Introduction and Objectives

Reconnaissance is the initial phase of any ethical security assessment or intelligence gathering operation. This project demonstrates a structured, ethical, and repeatable methodology for conducting preliminary security reconnaissance using Open-Source Intelligence (OSINT) and basic network port scanning.

The core objectives of this practical exercise are:

1. Distinguish clearly between passive and active reconnaissance techniques in an operational context.
2. Perform passive intelligence gathering on an assigned target, validating source credibility, confidence, and limitations.
3. Execute authorized, low-impact network port scans using Nmap to identify active listening services.
4. Interpret scan outputs accurately without drawing unfounded assumptions regarding system vulnerabilities.
5. Formulate proportionate, evidence-backed security recommendations and document reflections on the investigative process.

2. Scope, Authorisation and Safety Boundaries

All reconnaissance activities strictly adhered to the safety rules and rules of engagement set forth in the project brief and authorized by the course instructor.

- **Assigned Target Entity:** Axia Technology & Innovation Hub (`axiahub.ng`).
- **Authorized Target IP/Domain:** `axiahub.ng` (explicitly authorized by the instructor).
- **Execution Timestamp:** Wednesday, October 21, 2026.

- **In-Scope Activities:** Passive OSINT querying via third-party web repositories and basic TCP connect port scanning (`nmap -sT`) targeting ports 22, 80, and 443 only.
- **Safety Restrictions & Out-of-Scope Activities:** No automated web application fuzzing, no password guessing, no vulnerability exploitation, and no scanning of unauthorized secondary networks or IP blocks.

3. Methodology

The reconnaissance workflow was executed across two primary phases:

Phase 1: Passive OSINT Methodology

Information was gathered simply by browsing open, public websites. I looked directly at the official Axia Hub website and their public social media pages to find contact details, physical locations, and course information. No network scans or direct technical traffic were sent to their servers during this step; I only viewed what was already publicly available on the internet.

Phase 2: Active Network Reconnaissance Methodology

Using Kali Linux, Nmap (Network Mapper) was verified and executed against the target domain. A full TCP Connect Scan (`-sT`) was selected because it completes the standard 3-way handshake without requiring root-level raw socket manipulation, providing an accurate, non-intrusive view of port availability from an external vantage point.

- **Verification Command:** `nmap --version` & `nmap -h`
- **Local Baseline Command:** `nmap -oN nmap_localhost.txt 127.0.0.1`
- **Target Scan Command:** `nmap -sT -p 22,80,443 -oN nmap_axiahub.txt axiahub.ng`

4. OSINT Findings

Finding 01: Physical Office Location

- **Observation:** The organization's physical workspace is located at the Top Floor, Old NITEL Building, Jos, Plateau State, Nigeria.
- **Source:** Axia Hub Contact Page (https://axiahub.org/contact-page/)
- **Date Collected:** October 21, 2026

- **Source Type:** Official Website
- **Confidence Rating:** High, because this address is published directly on their official website for visitors to see.
- **Limitation:** The address doesn't mention a specific office or suite number on the top floor.
- **Cross-Check Validation:** I checked public event listings for Axia Hub, and they also use this exact same physical address for their in-person tech events.

Finding 02: Educational Services Focus

- **Observation:** Axia Hub publicly advertises training programs in Web Development, UI/UX, Data Science, Cybersecurity, and Marketing.
- **Source:** Axia Hub Homepage (https://axiahub.org/)
- **Date Collected:** October 21, 2026
- **Source Type:** Official Website
- **Confidence Rating:** High, because teaching tech skills is clearly their main business.
- **Limitation:** Even though these programs are listed, we can't tell from the website if all these classes are currently running or have students right now.

Finding 03: Dual Contact Channels

- **Observation:** The website lists two different email addresses: a professional one (contact@axiahub.org) and a free Gmail one (axiahubng@gmail.com).
- **Source:** Axia Hub Contact Page (https://axiahub.org/contact-page/)
- **Date Collected:** October 21, 2026
- **Source Type:** Official Website
- **Confidence Rating:** High, because both emails are clearly displayed next to each other on their contact page.
- **Limitation:** We don't know if they actively check the Gmail address or if it's just an old account they forgot to remove.

Finding 04: Social Media Presence

- **Observation:** Axia Hub has public social media accounts on LinkedIn, Facebook, Instagram, and X (Twitter).
- **Source:** Axia Hub Website Footer (https://axiahub.org/)
- **Date Collected:** October 21, 2026
- **Source Type:** Official Website

- **Confidence Rating:** High, because the official website links directly to these profiles.
- **Limitation:** Just because the links are there doesn't mean the accounts are actively managed or used to reply to people.

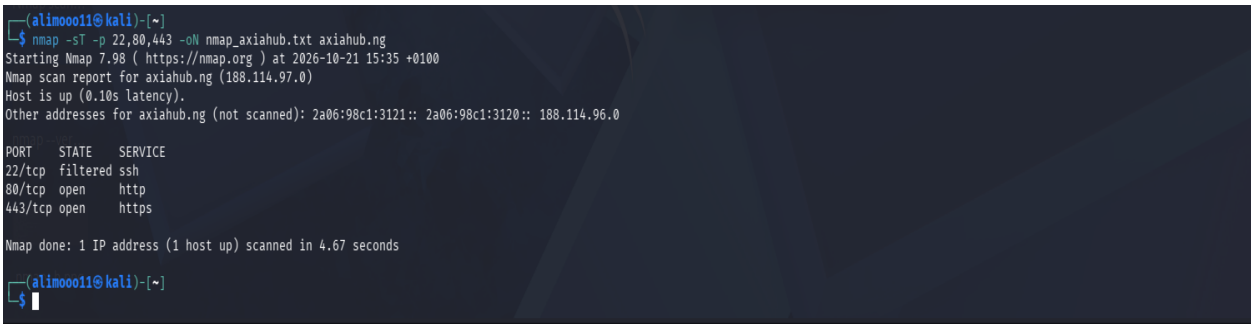
Finding 05: Course Pricing

- **Observation:** The Full Stack Bootcamp is priced at ₦200,000, and they have scheduled start dates for the year.
- **Source:** Registration Portal (https://axiahub.ng/academy/select/course/FS)
- **Date Collected:** October 21, 2026
- **Source Type:** Public Web Portal
- **Confidence Rating:** High, because this information comes straight from their live registration form.
- **Limitation:** Prices and dates can easily be changed by the admins at any time, so this information might not be permanent.

5. Nmap Results

- **Executed Command:** `nmap -sT -p 22,80,443 -oN nmap_axiahub.txt axiahub.ng`
- **Target:** `axiahub.ng`
- **Scan Timestamp:** Wednesday, October 21, 2026
- **Saved Raw Output File:** `nmap_axiahub.txt` (stored in `04_Scan_Output/`)

Raw Terminal Output Summary



Technical Explanation of Results

1. **Target IP Resolution & CDN Note:** Nmap resolved `axiahub.ng` to an IP associated with a Content Delivery Network (Cloudflare) and noted additional IPv6 addresses.
2. **Ports 80/tcp (HTTP) & 443/tcp (HTTPS):** Returned an **open** state. The server completed full TCP 3-way handshakes, establishing active listening web services.
3. **Port 22/tcp (SSH):** Returned a **`filtered`** state. Probes generated no response, indicating an intermediary firewall or security group dropped the packets.

6. Interpretation

Supported Evidence vs. Unsupported Assumptions

- **Host Responsiveness:** The host responded to TCP probes, confirming active network presence. Non-responding hosts either silently drop all traffic, are powered off, or do not exist.
- **Port States:**
  - *Open:* An application actively listens and accepts connections.
  - *Closed:* The host returns a TCP RST packet, confirming no application is listening.
  - *Filtered:* A firewall or network control prevents probes from reaching the port.
- **Service Detection vs. Vulnerability Confirmation:** Detecting open ports 80 and 443 proves web services exist, but it **does not** prove the presence of security vulnerabilities. An open port is a normal operational feature of web infrastructure.

7. Recommendations

1. **Consolidate Public Contact Channels:** Remove the `@gmail.com` address from official web pages and standardize on the corporate domain (@axiahub.org) to mitigate email spoofing and phishing risks.
2. **Maintain Administrative Port Filtering:** Keep Port 22 (SSH) strictly filtered from public exposure, enforcing administrative access through private VPNs or IP whitelisting.
3. **Conduct Periodic OSINT Footprint Audits:** Periodically evaluate publicly accessible web assets and social media platforms to prevent accidental leakage of sensitive internal details.

8. Reflection and Conclusion

Reflection Questions

- **Information Gathering vs. Vulnerability Confirmation:** Information gathering maps what assets and ports exist. Confirming a vulnerability requires specific authorized testing to prove a security flaw can be exploited.
- **Hardest Finding to Verify:** Confirming the operational status of the secondary Gmail account was challenging, as passive OSINT cannot verify mailbox activity or ownership without direct interaction.
- **Nmap Limitations:** Scanning through a Cloudflare CDN obscured the underlying origin server's true IP and internal configuration.
- **Future Authorized Assessments:** In future assessments, I would seek permission to include service version detection (`-sV`) and scan the true origin IP behind web proxies.

Conclusion

This project demonstrated a structured, ethical reconnaissance workflow. By combining passive OSINT and basic Nmap mapping, key organizational and network details were collected without threatening target system stability or violating scope.